



MKGW-mini Series BLE to WIFI Gateway

Product Specification

Version V1.4

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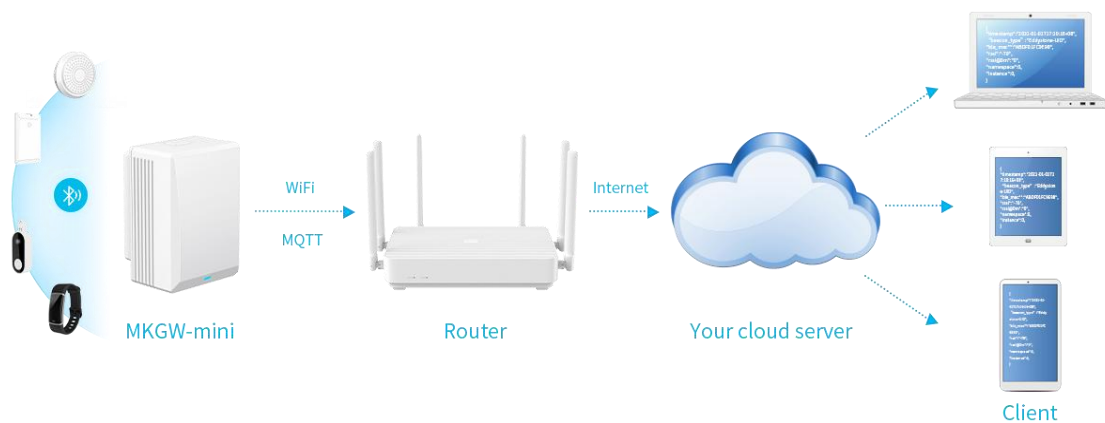
1. Product Introduction

1.1 Overview

MKGW-mini series product is a BLE (Bluetooth Low Energy) to WIFI gateway, which works as a data bridge between your beacon and cloud server. It scans and collects the advertising data of the surrounding beacon through Bluetooth, and uploads the Bluetooth data packet to your server through the WIFI network, effectively realize indoor positioning service, centralized asset tracking and real-time status monitoring in a low-cost way.

MKGW-mini comes equipped with interchangeable AC plugs and a Micro USB interface, supports flexible power supply methods. And it is simple to install, can be used in many countries.

MKGW-mini gateway works in MQTT protocol, supports connectivity with standard MQTT brokers, also can work with AWS iot and Alibaba iot cloud. For better data security, the gateway directly uploads data to the custom server, convenient for the further application development.



1.2 Product Model

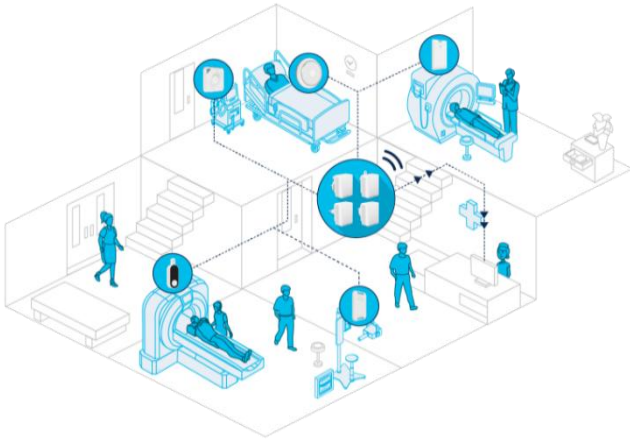
The MKGW-mini series includes several product models, see the below list to know the differences among these models.

Product Model	Description	Status
MKGW-mini 01-32	➤ 2.4GHz WIFI& BLE 4.2. ➤ WIFI security supports enterprise level (802.1X) ➤ Supports connectivity with MOKO Panic Button	Mass production
MKGW-mini 02-32B	2.4GHz WIFI& BLE 5.0	Mass production (NRND)
MKGW-mini 02-32D	➤ 2.4GHz WIFI& BLE 5.0 ➤ WIFI security supports enterprise level (802.1X) ➤ Supports connectivity with MOKO Panic Button	Mass production (Mainly recommended)
MKGW-mini 03-20B	2.4GHz&5GHz WIFI& BLE 5.0	Mass production (NRND)
MKGW-mini 03-20D	➤ 2.4GHz &5GHz WIFI& BLE 5.0 ➤ Supports connectivity with MOKO Panic Button	Mass production (Mainly recommended)

2. Features

- Mini size and compact design
- Equipped with interchangeable AC plugs, can be flexibly used in many countries
- The US plug is equipped with a locating hole to fix the gateway on the socket to prevent accident falling off
- Flexible power supply method, supports AC 100-240V and DC Micro USB power supply
- Provides mobile APP to do quick configuration for custom server connectivity
- Multiple data filter methods to obtain target beacon data easily
- Filter out the duplicate data to effectively reduce server resources
- Locally decode iBeacon, Eddystone (UID/URL/TLM) and all MOKO beacon raw data before uploading
- FCC&ETL&CE&UKCA certified

3. Application



Scenario 1: Smart Healthcare

Deploy the gateway in the medical place, and deploy beacons on the valuable/frequently used medical equipment. The gateway scans the beacon advertising data and uploads it to the server. With the server positioning algorithm, users can obtain the real-time position of the instrument, so as to realize asset tracking and management.

When emergency situation requires a medical equipment, the required equipment can be located in time without affecting or delaying patient care.

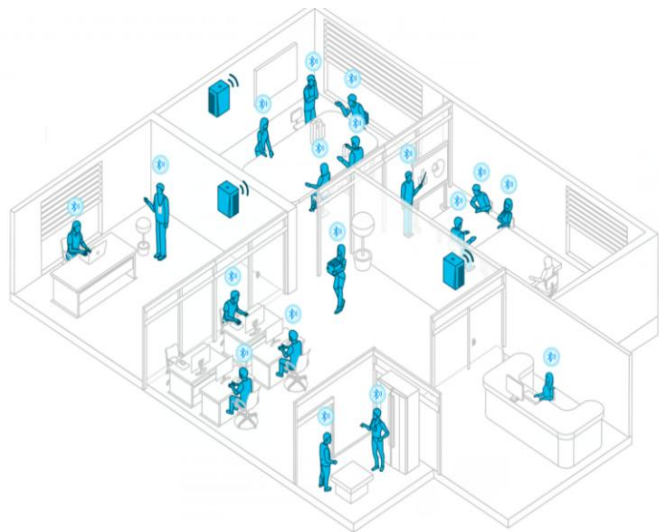
When an instrument is not detected for a period of time, a notification will be triggered, prompting the manager to find the device in time to reduce the risk of instrument loss.

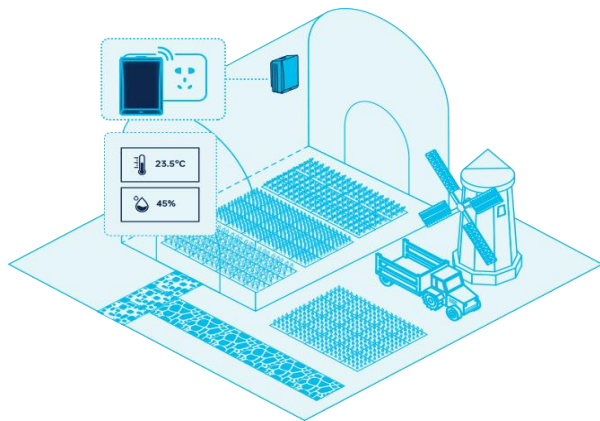
Scenario 2: Smart Workplace

In the smart office, the gateway only needs to scan the employee's ID card (Beacon) and upload the data to the server to easily realize intelligent attendance and employee interaction.

By deploying beacon on public office equipment, the gateway can also well track and manage office equipment, perform location query for public equipment, and reduce manual search time.

You can also know the occupancy of the meeting room in real time and arrange the meeting time reasonably.





Scenario 3: Smart Farm

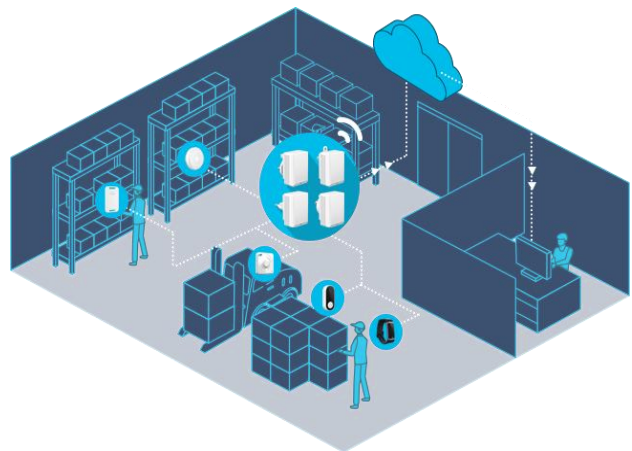
Deploying the gateway in the greenhouse and it can work with temperature and humidity sensors, easily realize real-time monitoring of the environment. Users can do data analysis on the server, generate temperature and humidity trend change reports, and understand hourly or even minutely environmental changes.

When there is an abnormal change in the environment, such as temperature/humidity exceeding the appropriate range, a notification will be triggered immediately, prompting the farmer to adjust the ambient

Scenario 4: Smart Factory

The gateway can be used in the smart factory to track important goods/equipment/vehicles. The real-time acceleration data of beacon indicates whether the equipment is moving or stationary at a certain location, and it can be judged whether the equipment is in use or not.

If a certain equipment has not been detected for a period of time, a notification will be sent to the manager in time, so as to search in time and reduce losses.



4. Product Appearance

4.1 Product Appearance

1. Main body
2. Indicator LED
3. Micro USB interface



4. Reset Button
5. Push Button



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6. Interchangeable UK plug
7. Interchangeable US plug
8. Interchangeable EU plug
9. Interchangeable AU plug

⑦

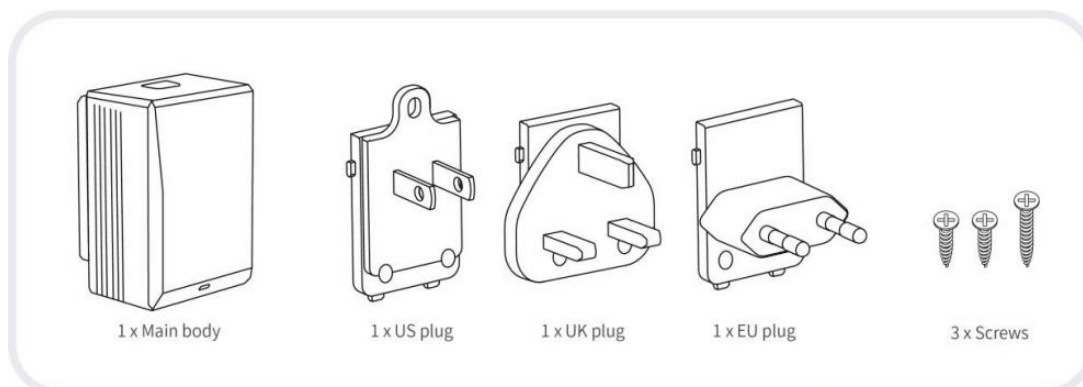


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4.2 Standard Package List



Please note: AU plug is not included in the standard package

5. Product Specification

5.1 Basic

Electronic	
Plug type	US/UK/EU/AU (AU plug is not included in the standard package)
Power supply	100-240VAC, 50/60Hz; 5V Micro USB
Button	1* button used for reset device
LED indicator	1* RGB indicator
Antenna	BLE and WIFI are both PCB antenna
Physical	
Material	ABS + PC
Color	White
Dimension	61.8mm*43.4mm*35mm
Environment	
Operating temperature	0 °C~ 40 °C
Operating humidity	0%~95% (No condensation)
Storage temperature	-10 °C~ 50 °C
Communication	
Protocol	MQTT V3.1.1
Encryption	TCP/SSL
Data format	JSON

5.2 Wireless

WIFI	MKGW-mini 01-32	MKGW-mini 02-32D	MKGW-mini 03-20D
Module/chip	ESP32-WROOM-32	ESP32-WROOM-32	RTL8720
Bandwidth	2.4GHz	2.4GHz	2.4GHz&5GHz
Protocol	802.11 b/g/n	802.11 b/g/n	802.11 a/b/g/n
Security	Personal&Enterprise(802.1X)	Personal&Enterprise(802.1X)	Personal

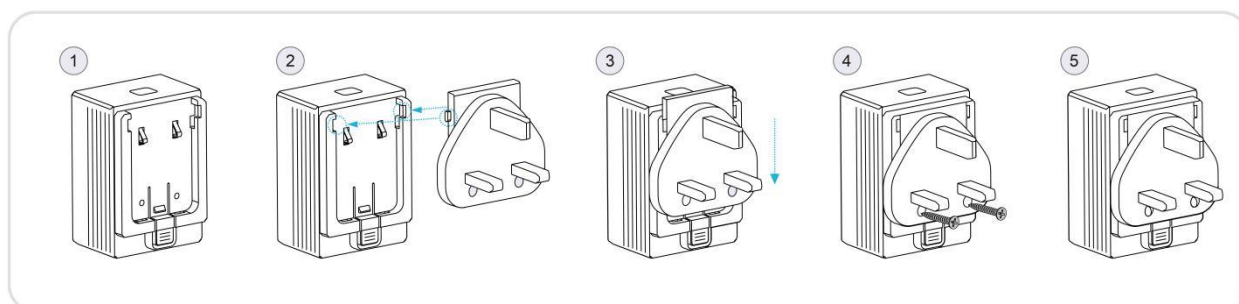
BLE	MKGW-mini 01-32	MKGW-mini 02-32D	MKGW-mini 03-20D
Module/chip	Same as WIFI module	ESP32-C3-MINI	ESP32-C3-MINI
Protocol	BLE 4.2	BLE 5.0	BLE 5.0
Scan range in open space (meters)	120	BLE 4.2: 150 BLE5.0(1M PHY): 150	BLE 4.2: 150 BLE5.0(1M PHY): 150
Scan rate (BLE packets/second)	118	260	220

6. User Guide

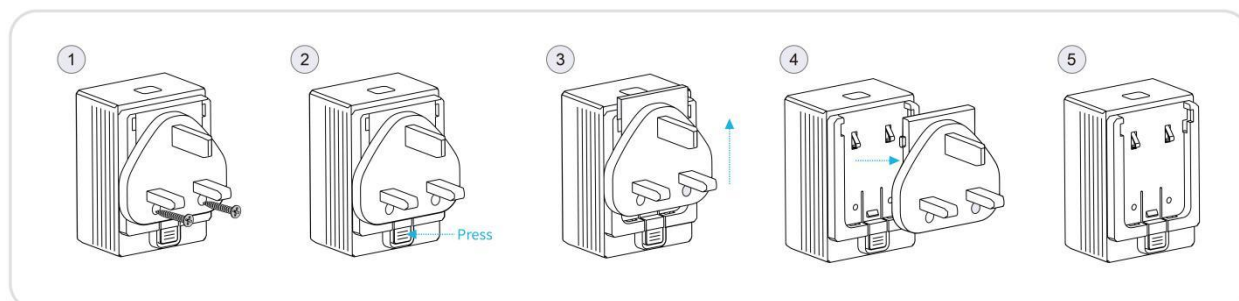
6.1 How to install/remove the AC Plug?

The MKGW-mini gateway is equipped with three interchangeable AC plugs, users can flexibly replace the AC plug to apply it in different countries.

How to install the AC plug?

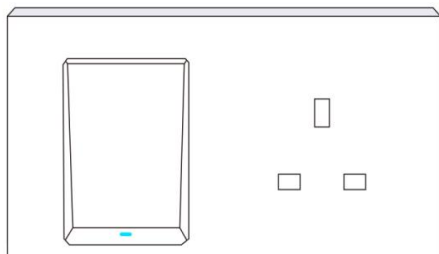


How to remove the AC plug?



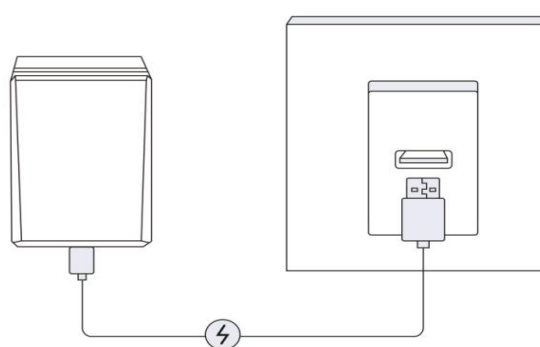
6.2 How to install the gateway?

The gateway will start working immediately after it is powered up. The gateway supports flexible power supply methods, which can be powered by AC100-240V or a Micro USB cable.



Option 1: Plug in the gateway into a power socket

Option 2: Use a Micro USB to power on the gateway



7. Main Function

7.1 Flexible configuration

When the gateway is tested for the first time, it will start BLE advertising after powered on, the advertisement can be scanned and connected by MOKO APP.

After the gateway Bluetooth is connected by the mobile APP, users can send WIFI and MQTT information to the gateway via the APP, and then the gateway will connect to the WIFI and MQTT broker.

7.2 Automatic re-connection

If the gateway disconnects from server due to the network issue or power losing, after the network or power supply come back, the gateway will automatically reconnect to the WIFI and server which the gateway is successfully connected before disconnection.

7.3 Gateway uplink

Gateway scans for nearby BLE devices and upload the BLE data to the cloud.

7.3.1 Scan BLE Devices

The gateway supports to collect iBeacon, Eddystone and all BLE 4.2, BLE 5.0 device data.

7.3.2 Data Filtering

The gateway supports multiple data filtering methods to help you easily obtain target beacon data:

- Filter by RSSI, MAC address, advertising name and advertising raw data.
- Filter out duplicate data, reports only non-duplicate data to your server in a filtering period.

7.3.3 Data Decoding

The gateway can locally decode the scanned Bluetooth advertising data before uploading to server. The users can directly obtain the desirable information on the server.

- Support standard iBeacon, Eddystone (UID/URL/TLM) raw data decoding.
- Support all MOKO beacon raw data decoding.

7.3.4 Data Upload

The gateway uploads Beacon packets to server through WiFi. The uploading data content is configurable, the users can configure the gateway reporting data to include only what they need, which will effectively save server and network resources.

7.4 Gateway downlink

Gateway connects and communicates with Beacon device. The users can remotely manage Beacon devices by sending command to the gateway from cloud.

The typical application is to use the gateway and MOKO B1/B2 panic button for personal security management solution. In hospital, the staffs wear the panic button, when an emergency happens, the staff presses the SOS button to trigger the alarm message advertising. The gateway transfers the alarm message to cloud, and connect to the panic button to inform that the alarm is received by the management team.

Please note: This feature is supported on the MKGW-mini 01, MKGW-mini 02-32D, MKGW-mini 03-20D gateways, not supported on MKGW-mini 02-32B and MKGW-mini 03-20B gateways.

7.5 OTA

The gateway has the ability to upgrade firmware over the air. MOKO can provide the latest firmware upgrade file with customers for their further test.

7.6 Reboot

The gateway supports software reboot via the cloud command.

7.7 Restore to Factory Settings

The gateway is equipped with a button for reset operation. Press the button and hold for 10 seconds, the gateway will restore factory settings, and then it will enter configuration mode.

The gateway also supports software reset via the cloud command.

8. Product documents

MOKO provides the following documents to help customers test the gateway, do platform integration based on MOKO standard firmware, and develop custom firmware.

File	Description
<i>User Manual</i>	This document instructs users how to configure the gateway with MOKO APP and get BLE data from Beacons. It also include the APP SDK, which is the APP source code.
<i>Communication Protocol</i>	This document describes the configuring commands and management commands supported on gateway
<i>Firmware Development Document</i>	The document includes the schematic diagram and test points description which is related to firmware development and download. With this document, customers can develop their own firmware and download the firmware to the MOKO hardware.

Revision History

Revision	Description	Editor	Date
V1.0	Initial Version	Weiguifen	2021.8.28
V1.1	Add series models, and update product pictures.	Weiguifen	2022.3.25
V1.2	Update product certification, package list and product model description	Weiguifen	2022.8.23
V1.3	Update product models, product specification and main function	Weiguifen	2024.3.12
V1.4	Add AU plug, update product models and product specification	Weiguifen	2024.5.9

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